



IC 136: Understanding Biotechnology & its Applications

Course Code	IC 136
Course Name	Understanding Biotechnology & its Applications
L-T-P-C	3-0-0-3
Prerequisites	Consent of the faculty member
Intended for	B.Tech
Distribution	Core
Approval	2nd Adhoc Senate (May 2012), in the B.Tech Institute Core (common core courses) curriculum

Course Contents

- Introduction to “biotechnology” and the history of biotechnological developments with major milestones. [1 Lecture]
- Basic biology: Brief introduction to genes and genomes. [3 Lectures]
- Introduction to recombinant DNA technology and its application to genomics.[5 Lectures]
- Introduction to proteins and their products. [4 Lectures]
- Microbial biotechnology. [5 Lectures]
- Plant biotechnology. [5 Lectures]
- Animal biotechnology. [5 Lectures]
- Bioremediation and environmental biotechnology. [5 Lectures]
- Medical biotechnology. [5 Lectures]
- Biotechnology regulations and ethics. [2 Lectures]

Text Books

1. William J. Thieman and Michael A. Palladino, **Introduction to Biotechnology**, 3rd Edition, Benjamin-Cummings publishing company.

References

1. Reinhard Renneberg, **Biotechnology for Beginners**, Academic press.
2. Ratledge Colin, **Basic Biotechnology**, 3rd Edition, ambridge university press.